Illumina Rapid RNA vs Watchmaker RNA library prep for mNGS

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2026

Sequencing workflows are made up of individual modules

Collection and Transport

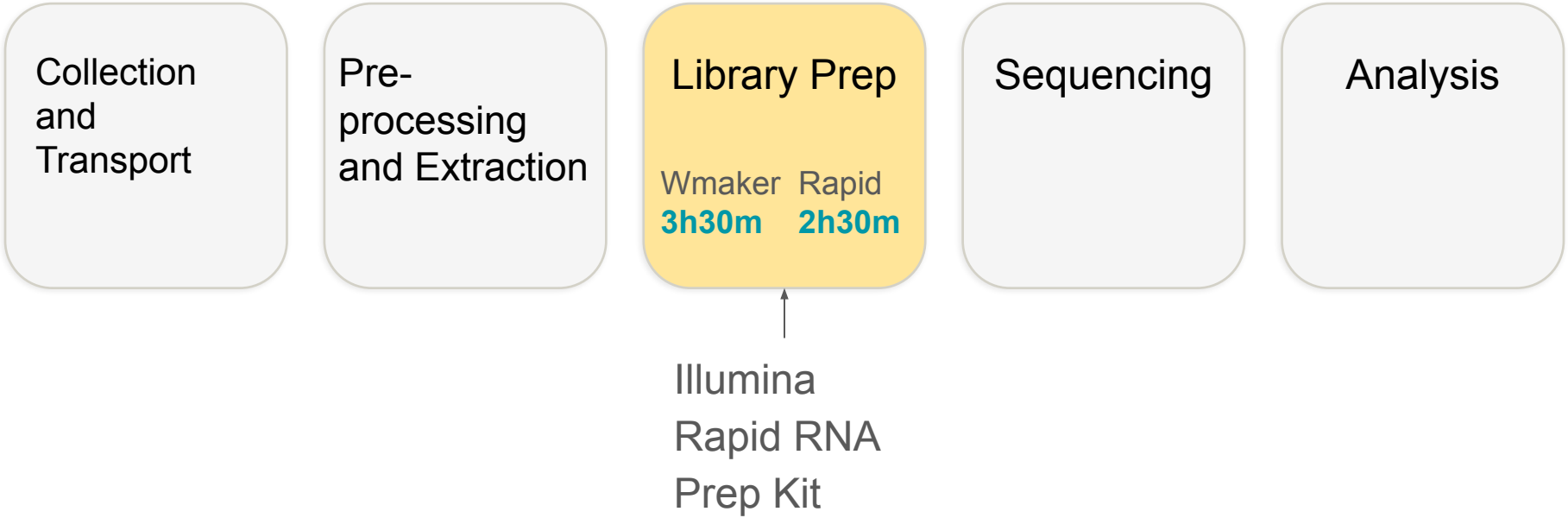
Pre-processing and Extraction

Library Prep

Sequencing

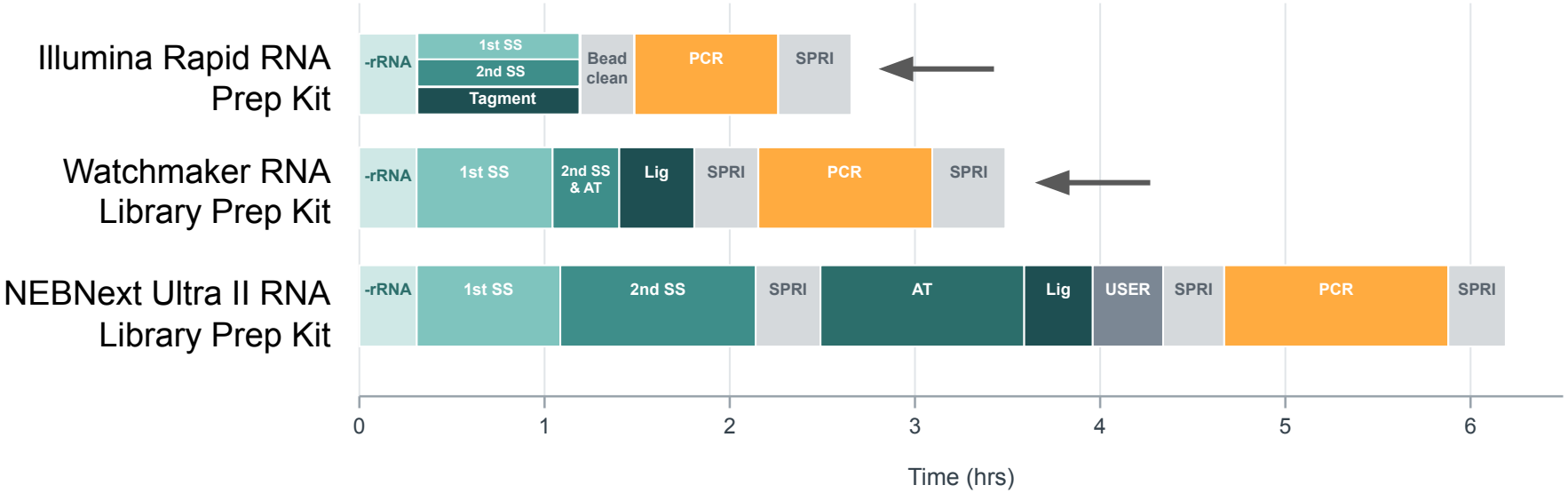
Analysis

Adjusting individual modules improves turnaround time



Goal: evaluate the new Illumina Rapid RNA Prep Kit for RNA mNGS library prep

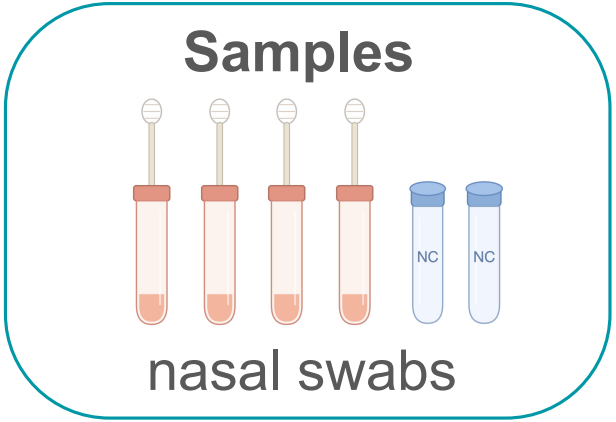
Methods: Illumina Rapid RNA Prep is a more streamlined workflow than Watchmaker RNA prep or NEBNext Ultra II RNA



Samples: Nasal swab RNA was used to compare the two kits

Samples

- 4 virus-positive nasal swabs
- 2 negative water controls



Kits

- Watchmaker RNA + FastSelect
- Illumina Rapid RNA Prep + FastSelect
- Illumina Rapid RNA Prep, no FastSelect

Modifications

+ ERCC synthetic RNA internal positive controls
+ Qiagen FastSelect to deplete human rRNA (1:10 diluted)

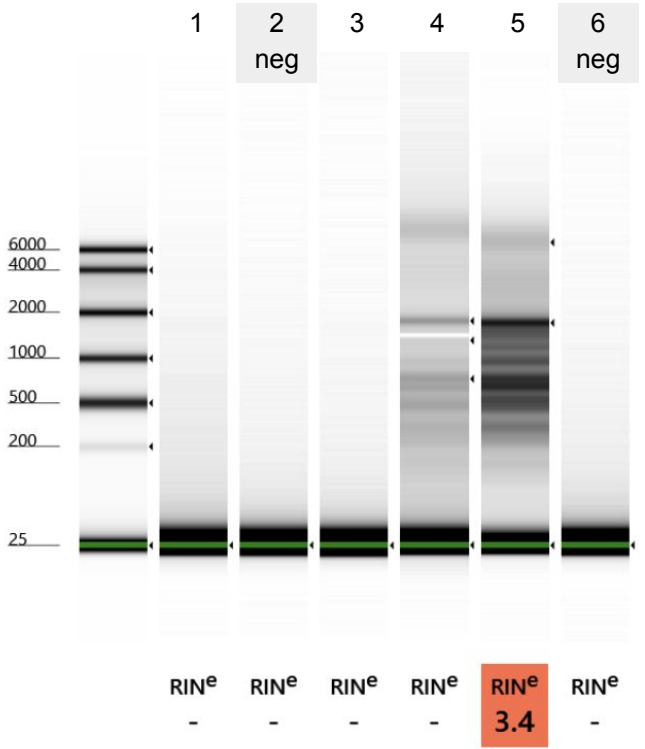
Watchmaker has potential for more RNA input volume than Illumina Rapid RNA

Protocol max input amounts				
Kit	**uL RNA input (protocol)**	**uL Sample RNA**	**uL ERCC control**	**uL FastSelect**
Watchmaker + FastSelect	15	13.5	0.5	1
Illumina Rapid + FastSelect	8.5	7	0.5	1
Illumina Rapid, no FS	8.5	8	0.5	-

For low input samples, higher sample input volume can be advantageous

Equal amounts of RNA was input to each protocol

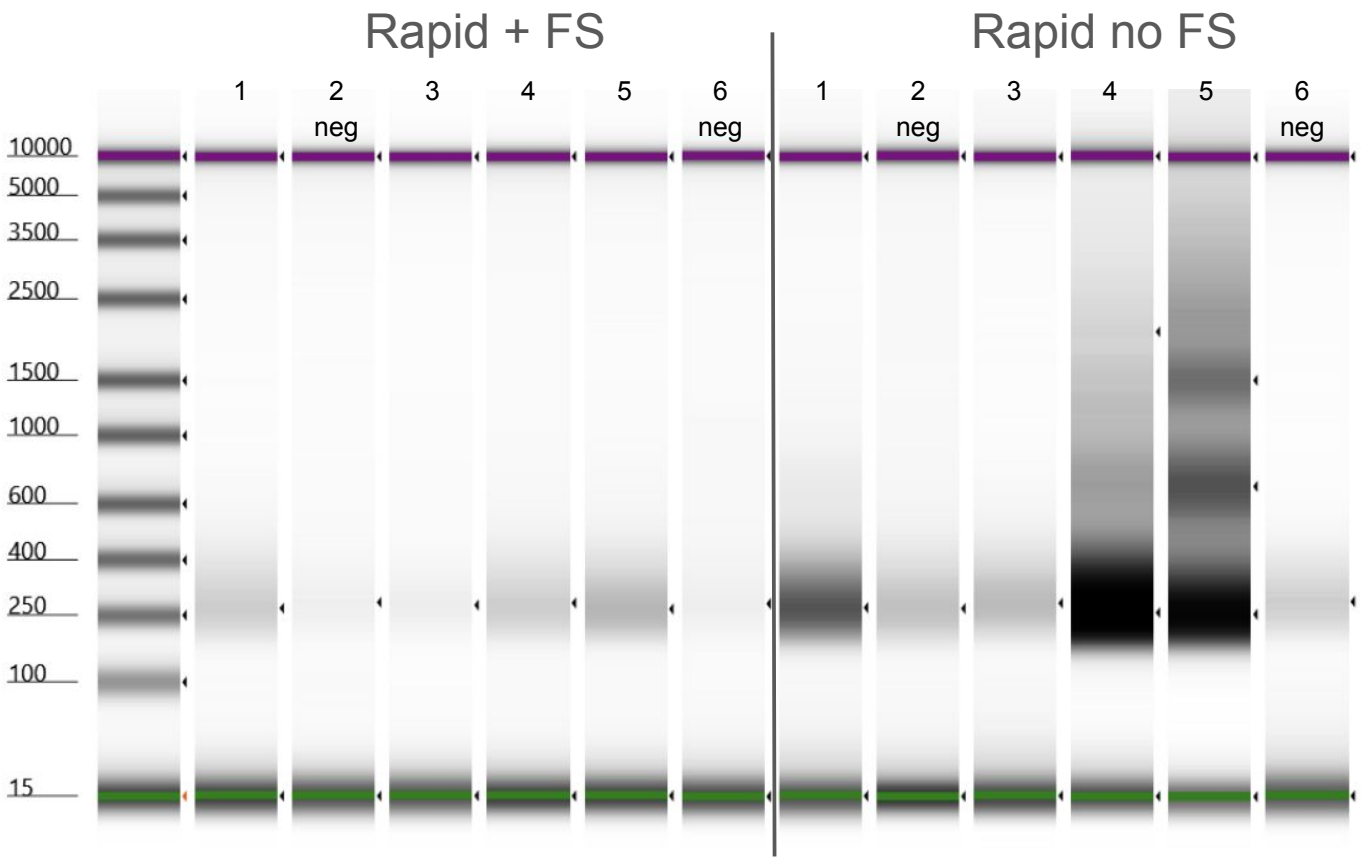
5ul RNA input per sample		
Sample	**ng input RNA**	**ng/ul RNA**
1	0.25	0.050
2 neg	0.165	0.033
3	0.145	0.029
4	0.865	0.173
5	2.295	0.459
6 neg	0.16	0.032



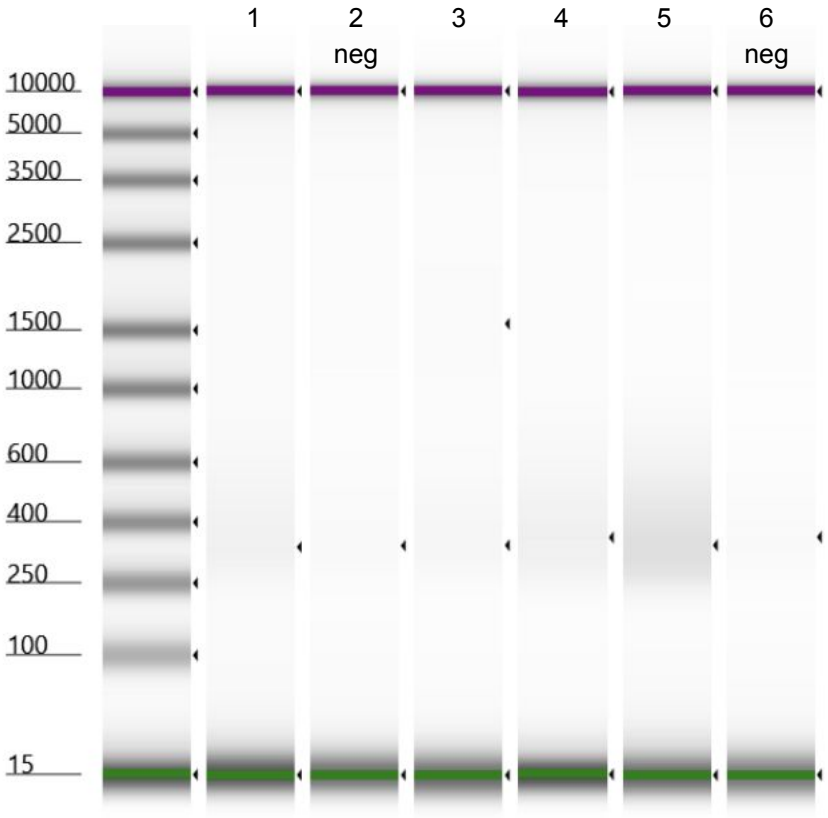
Library yield was good in all cases; highest without FastSelect

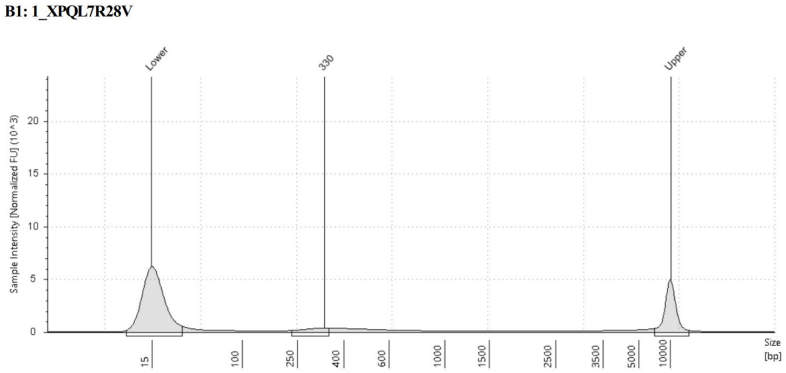
ng/ul of completed libraries (Qubit)			
	Watchmaker + FS	**Rapid + FS**	**Rapid no FS**
PCR Cycles:	**12**	**15**	**15**
1	1.02	3.3	15.2
2 neg	0.40	0.79	3.52
3	0.63	0.98	5
4	1.52	3.2	56
5	3.62	5.36	85.6
6 neg	0.43	0.71	3.42

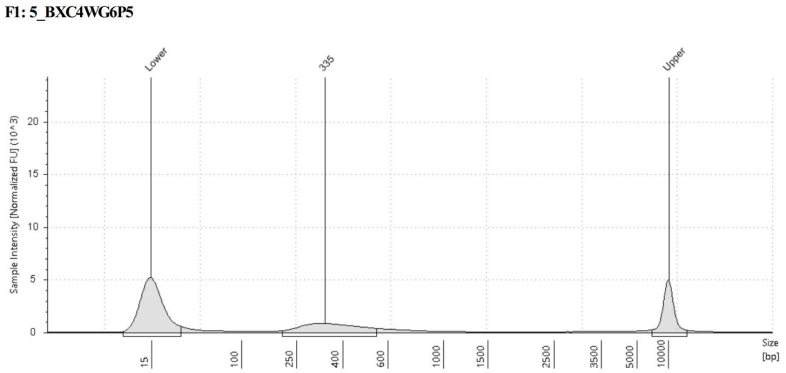
Library QC: Rapid Libraries +/- FastSelect are of good quality



Library QC: Watchmaker Libraries are good quality





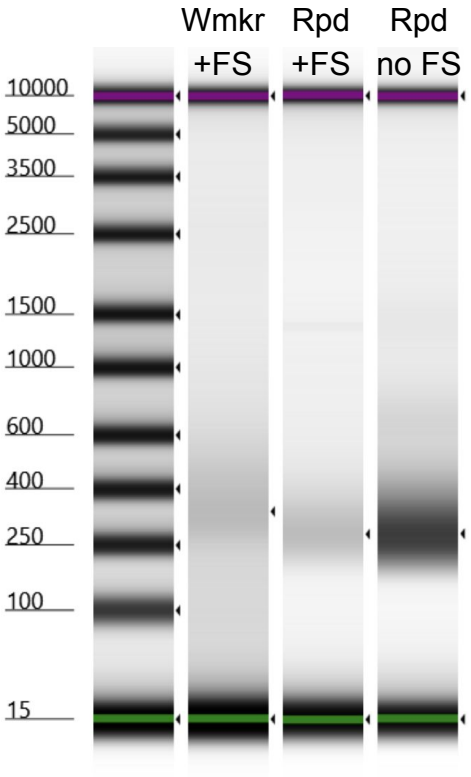


*Concentrations are slightly lower because fewer PCR cycles were completed

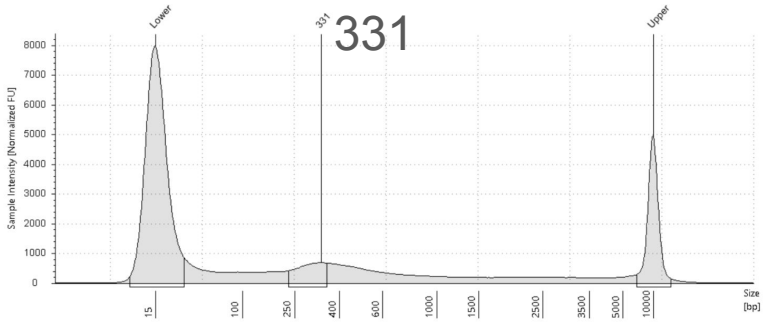
PCR cycles, library size, and sequencing reads differed between protocols

Workflow comparisons			
Step	**Watchmaker + FS**	**Rapid + FS**	**Rapid no FS**
Input RNA	5ul	5ul	5ul
FastSelect	1:10	1:10	none
PCR cycles	12	15	15
Library QC Mean lib size	331bp	274bp	275bp
Reads per sample	low	high	high

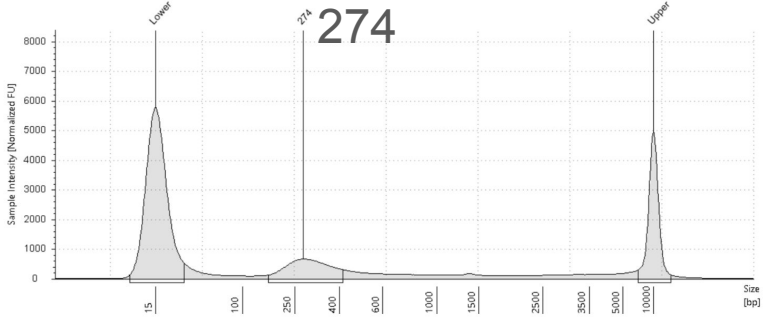
Watchmaker pool is slightly larger with more background



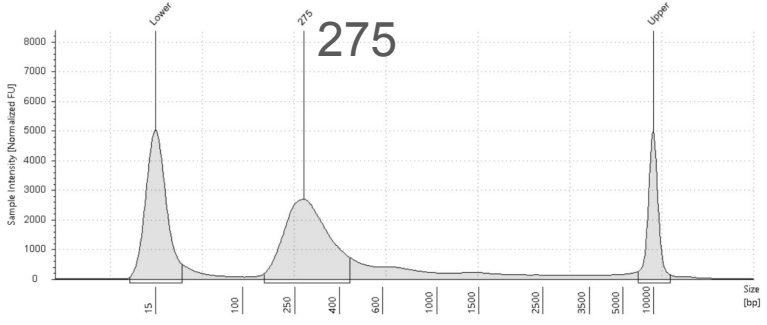
Wmkr +FS



Rpd + FS



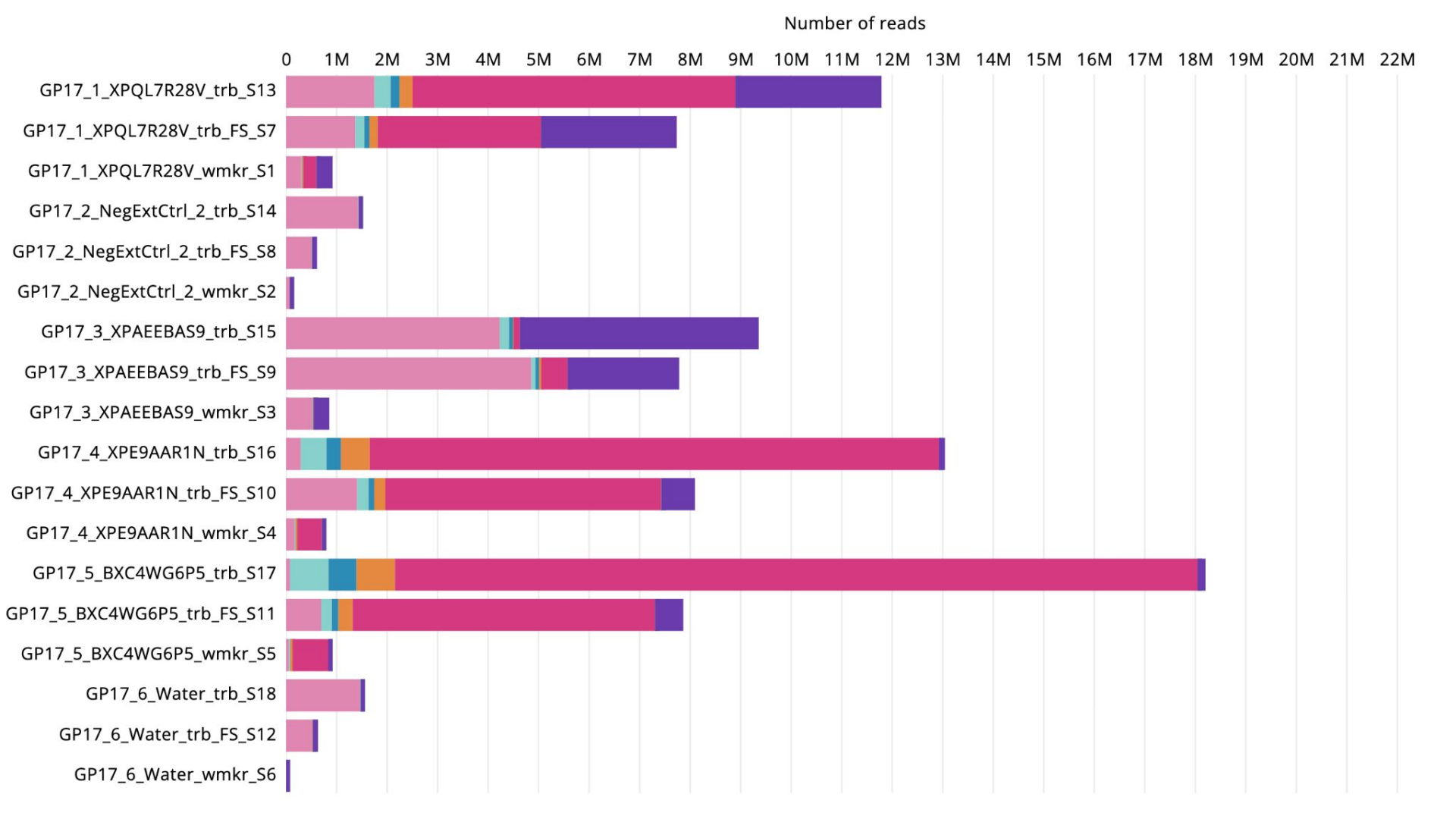
Rpd no FS



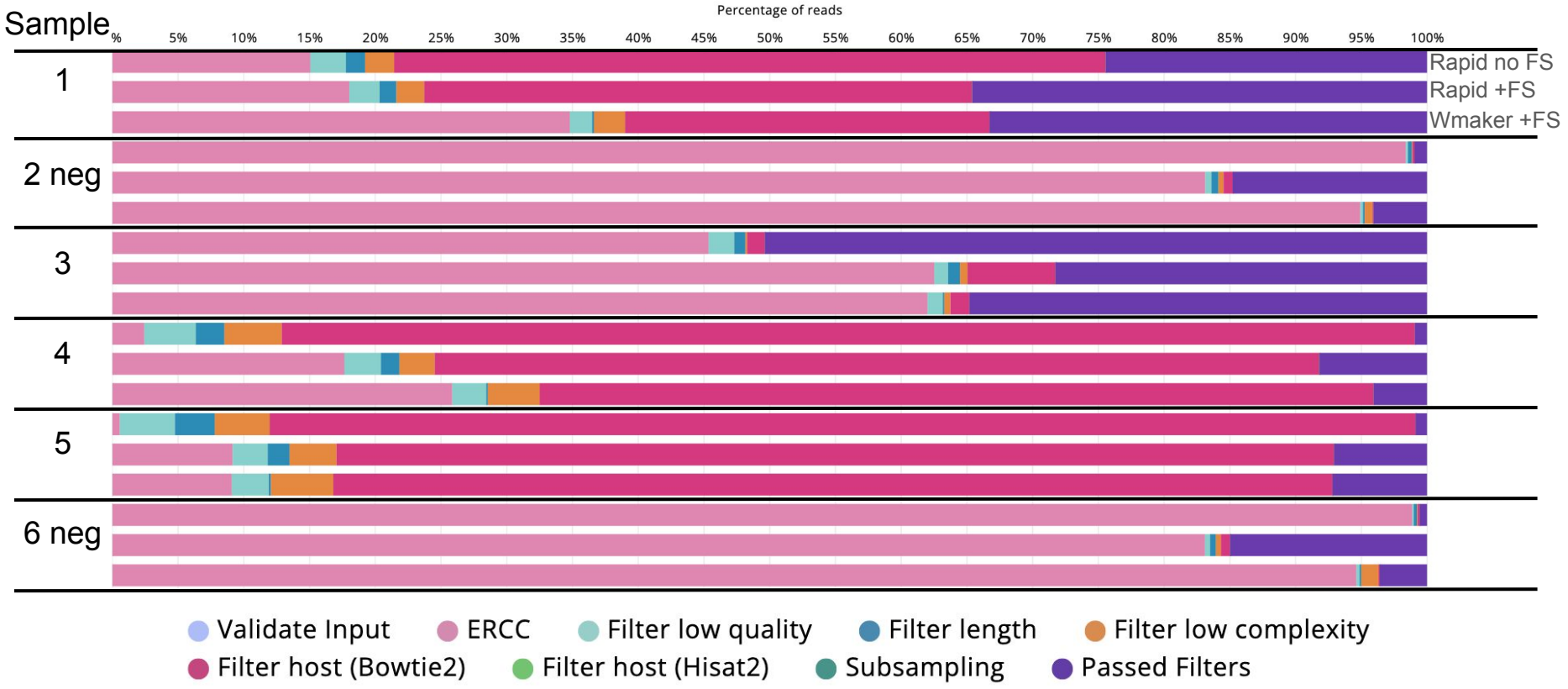
Watchmaker+FS samples received only ~11% of the Rapid+FS sample reads

Average total reads per sample		
Kit	**Samples (n=4)**	**Negatives (n=2)**
Rapid no FS	13M	1.5M
Rapid +FS	7.9M	0.65M
Watchmaker +FS	0.88M	0.06M

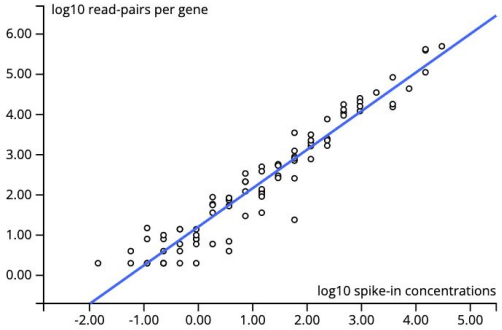
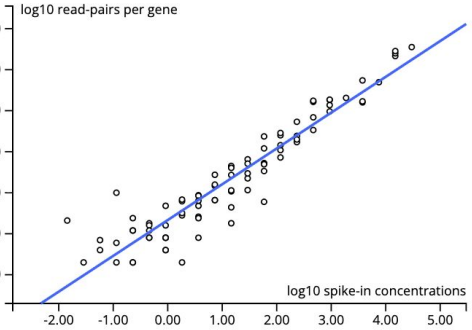
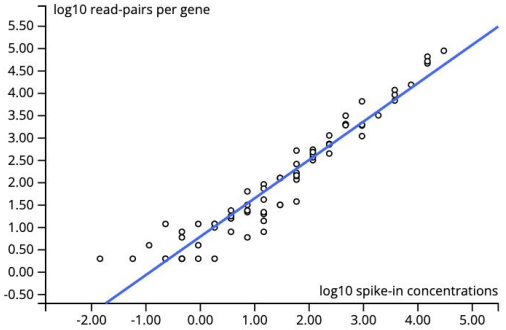
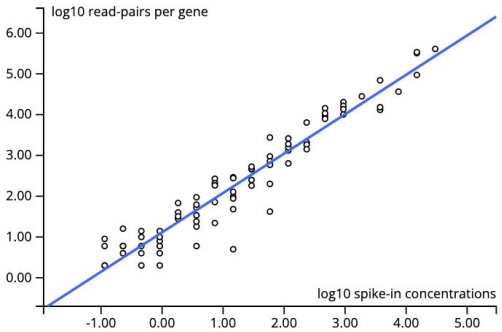
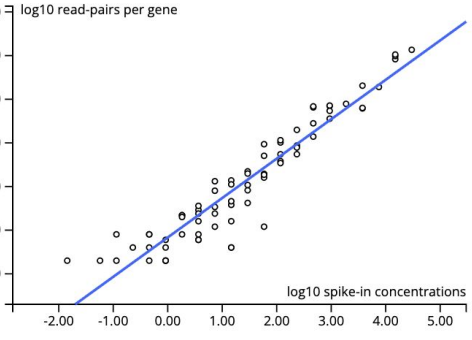
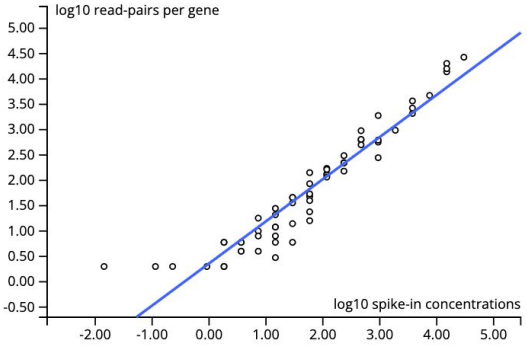
* Pooling could be improved
* Watchmaker could be amplified another cycle


Number of reads
0
1M
2M
3M
4M
5M
6M
7M
8M
9M
10M
11M
12M
13M
14M
15M
16M
17M
18M
19M
20M
21M
22M
GP17_1_XPQL7R28V_trb_S13
GP17_1_XPQL7R28V_trb_FS_S7
GP17_1_XPQL7R28V_wmkr_S1
GP17_2_NegExtCtrl_2_trb_S14
GP17_2_NegExtCtrl_2_trb_FS_S8
GP17_2_NegExtCtrl_2_wmkr_S2
GP17_3_XPAEEBAS9_trb_S15
GP17_3_XPAEEBAS9_trb_FS_S9
GP17_3_XPAEEBAS9_wmkr_S3
GP17_4_XPE9AAR1N_trb_S16
GP17_4_XPE9AAR1N_trb_FS_S10
GP17_4_XPE9AAR1N_wmkr_S4
GP17_5_BXC4WG6P5_trb_S17
GP17_5_BXC4WG6P5_trb_FS_S11
GP17_5_BXC4WG6P5_wmkr_S5
GP17_6_Water_trb_S18
GP17_6_Water_trb_FS_S12
GP17_6_Water_wmkr_S6

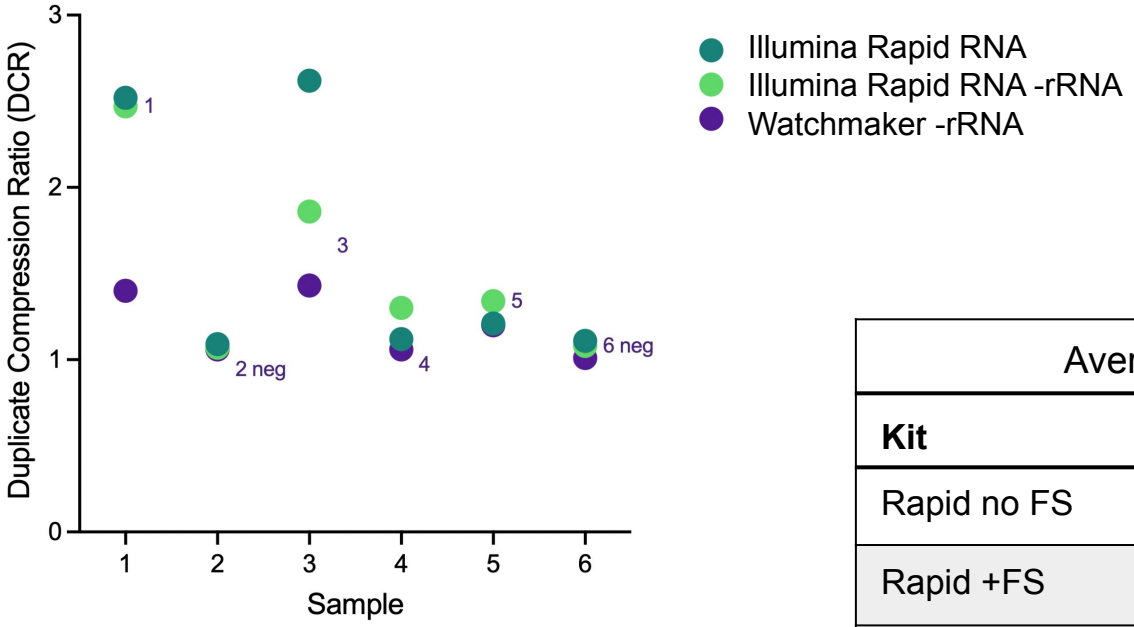
Reads QC indicates high quality libraries from all preps; Watchmaker +FS and Rapid +FS have similar composition



Internal positive controls (ERCCs) look very good with all preps

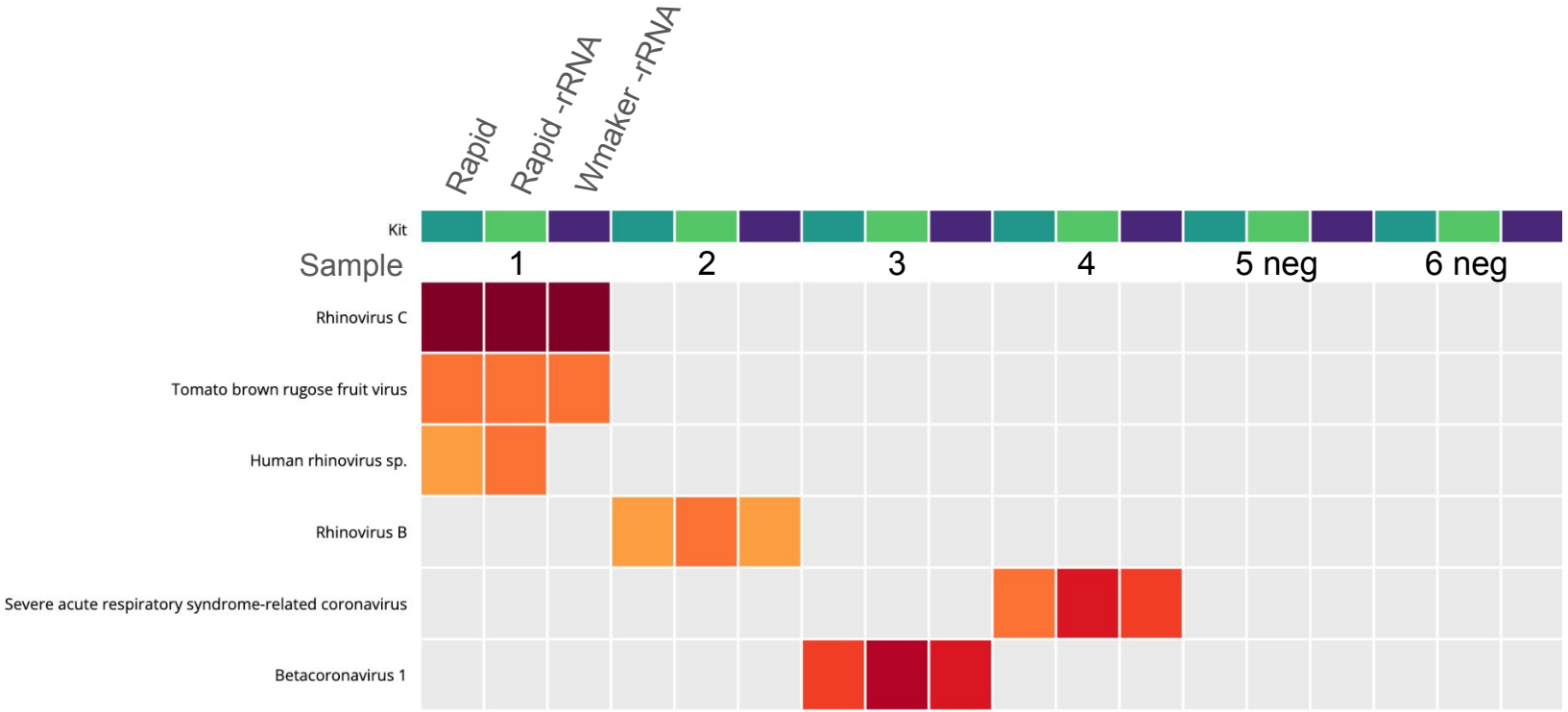
	Rapid no FS	Rapid +FS	Watchmaker +FS
1			
2 neg			

Duplicate Compression Ratios are all < 3 (good!), however comparisons are confounded by differences in total reads



Average total reads per sample		
Kit	**Samples (n=4)**	**Negatives (n=2)**
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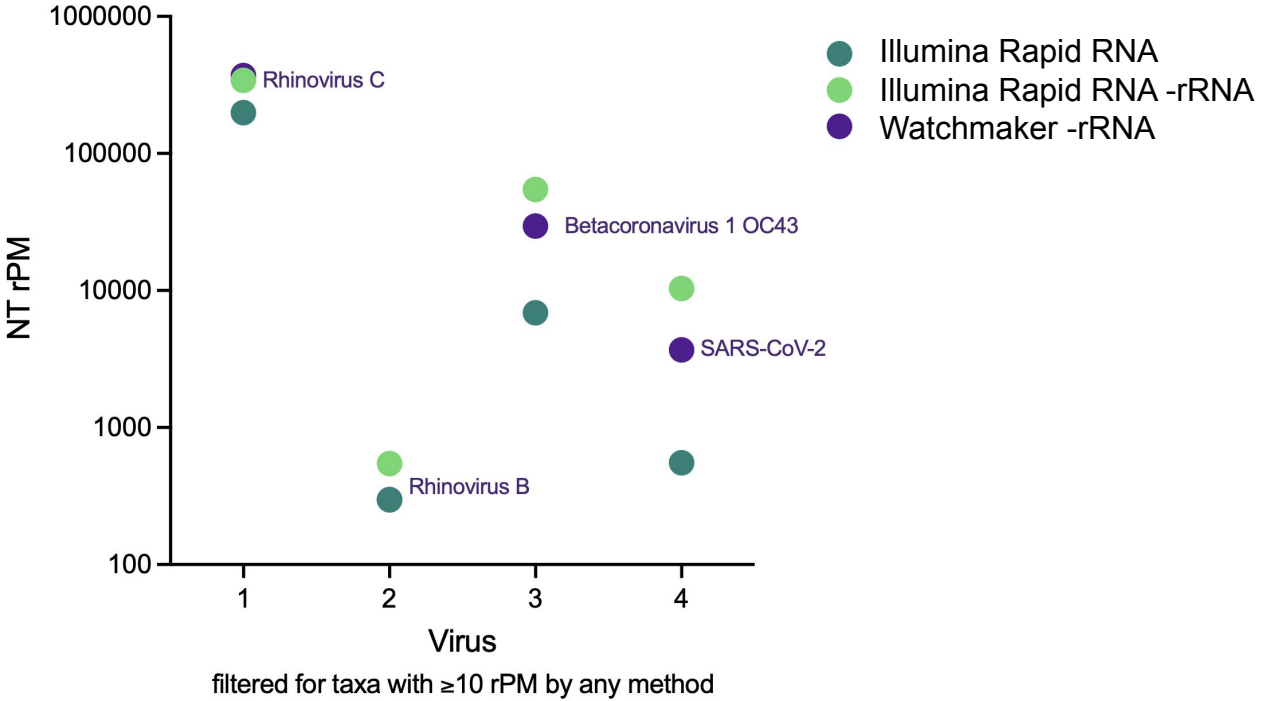
Virus **detection** is similar between prep type; **relative abundance** is lowest without -rRNA FastSelect



Threshold filters

Viruses; NT rPM ≥ 10; NT r ≥ 10

Relative abundance of viruses is improved with human rRNA depletion with FastSelect



High abundance Rapid hits have more (but probably shorter) contigs than Watchmaker hits

Rapid + FS

Taxon	Score	Z Score	rPM	r	co...	co...	%id	L	E ...
Enterovirus (*11 viral species:* ● 5)	-	-	**342,392.5** 296,417.6	**2,178,622** 1,886,087	**2,144** 919	**1,559,597** 1,265,101	**96.1** 96.2	**174.0** 53.1	**10^{-61}** 10^{-27}
Rhinovirus C **Known Pathogen**	-	-	**341,701.7** 279,463.2	**2,174,227** 1,778,207	**2,134** 870	**1,555,249** 1,198,427	**96.1** 96.7	**174.1** 54.0	**10^{-61}** 10^{-28}
Human rhinovirus sp. **Known Pathogen**	-	-	**679.9** 11,457.6	**4,326** 72,904	**9** 33	**4,309** 38,559	**94.4** 90.3	**100.3** 38.4	**10^{-30}** 10^{-16}
Rhinovirus A **Known Pathogen**	-	-	**9.4** 3,223.4	**60** 20,510	**1** 4	**39** 13,716	**98.4** 70.5	**84.4** 35.0	**10^{-18}** 10^{-9}

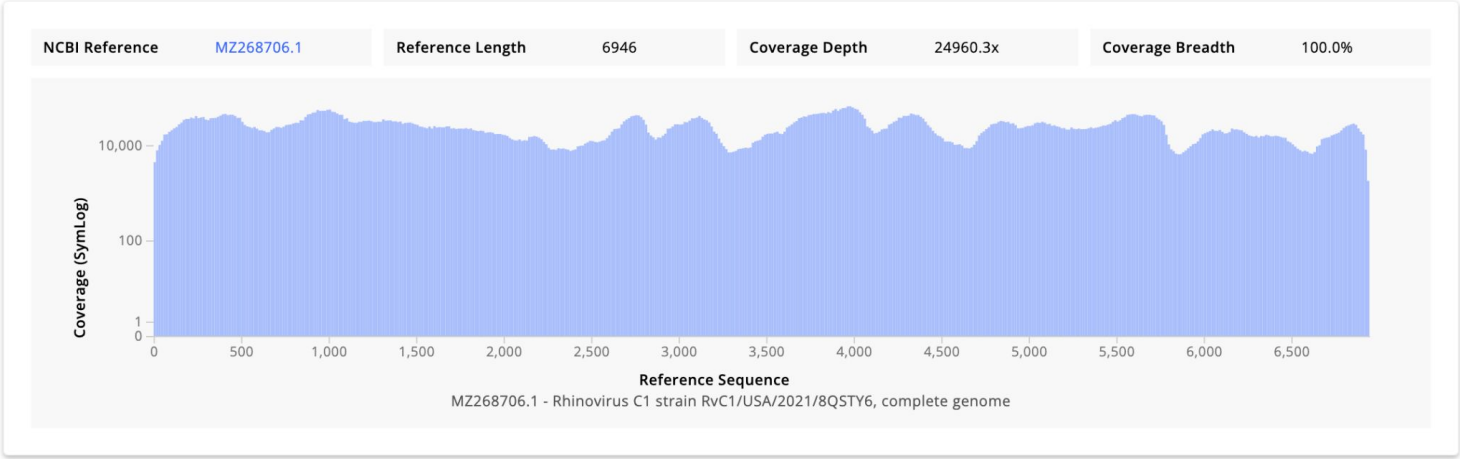
Watchmaker +FS

Taxon	Score	Z Score	rPM	r	co...	co...	%id	L	E ...
Enterovirus (*2 viral species:* ● 2)	-	-	**370,004.0** 369,979.8	**228,570** 228,555	**1** 1	**228,375** 228,375	**92.5** 97.7	**6,926.9** 2,141.7	**10^{-307}** 10^{-307}
Rhinovirus C **Known Pathogen**	-	-	**370,004.0** 369,976.5	**228,570** 228,553	**1** 1	**228,375** 228,375	**92.5** 97.7	**6,926.9** 2,141.7	**10^{-307}** 10^{-307}
Human rhinovirus sp. **Known Pathogen**	-	-	**0.0** 3.2	**0** 2	**0** 0	**0** 0	**0.0** 80.5	**0.0** 36.0	**0** 10^{-8}

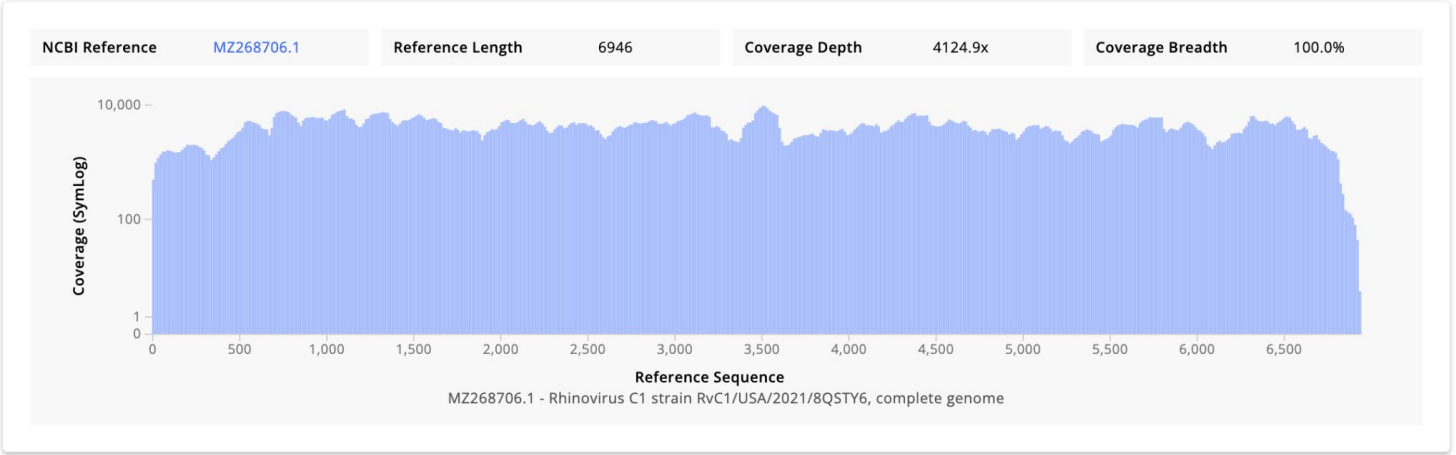
* The reason for high number of contigs despite complete coverage breadth (lab workflow vs analysis pipeline) is still under investigation

High titer viral consensus genomes both have full coverage

Illumina Rapid -rRNA



Watchmaker -rRNA



Let’s look at a low abundant virus (Rhino B, Sample 3)

Rapid + FS

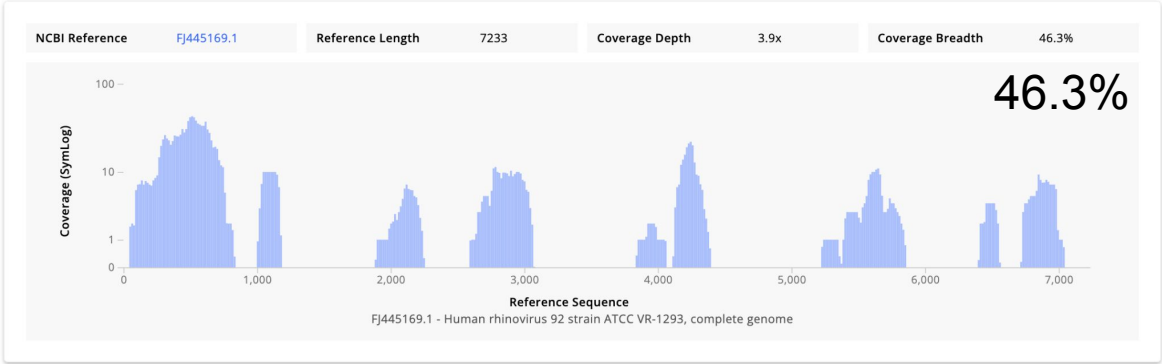
Taxon	Score	Z Score	rPM	r	co...
Enterovirus (*2 viral species:* ● 1)	-	-	**548.5** 550.6	**1,606** 1,612	**6** 6
Rhinovirus B	-	-	**546.5** 549.9	**1,600** 1,610	**6** 6
Rhinovirus C Known Pathogen	-	-	**2.0** 0.7	**6** 2	**0** 0

Watchmaker +FS

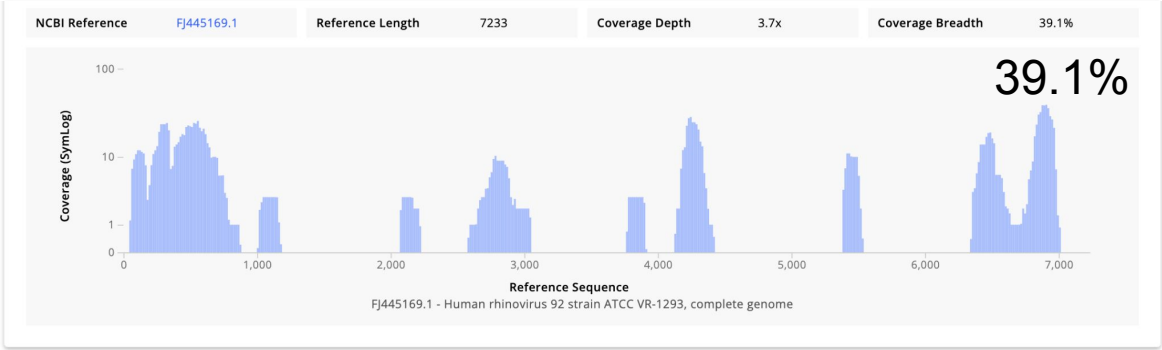
Taxon	Score	Z Score	rPM	r	co...
Enterovirus (*1 viral species*)	-	-	**297.8** 262.1	**100** 88	**10** 10
Rhinovirus B	-	-	**297.8** 262.1	**100** 88	**10** 10

Coverage of a low abundant virus

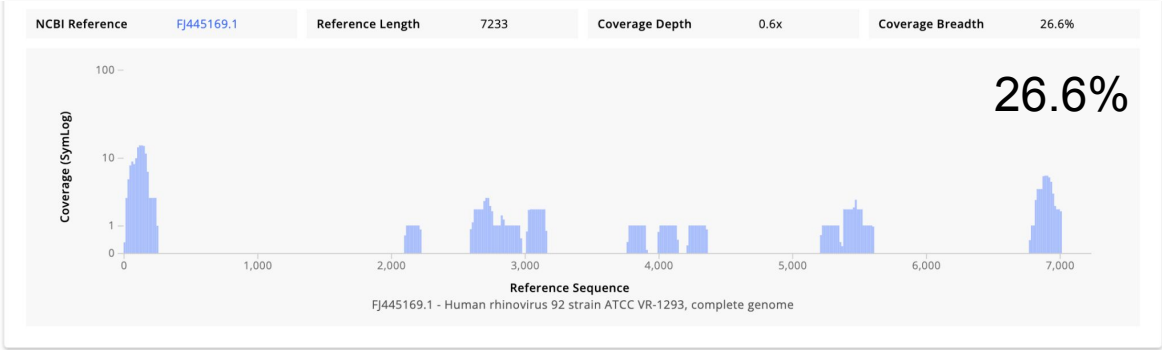
Illumina Rapid
7,435,518 total reads



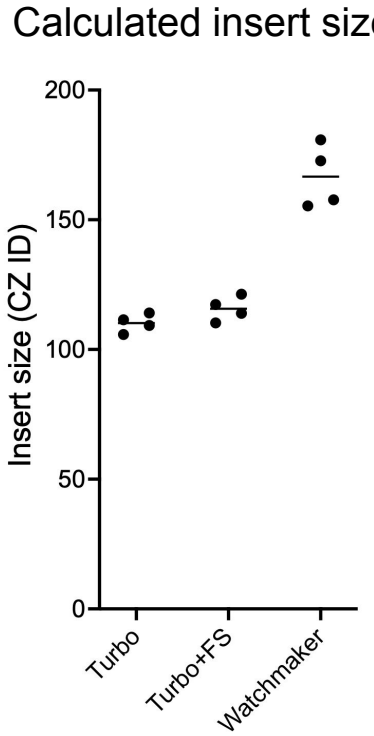
Illumina Rapid -rRNA
7,810,918 total reads

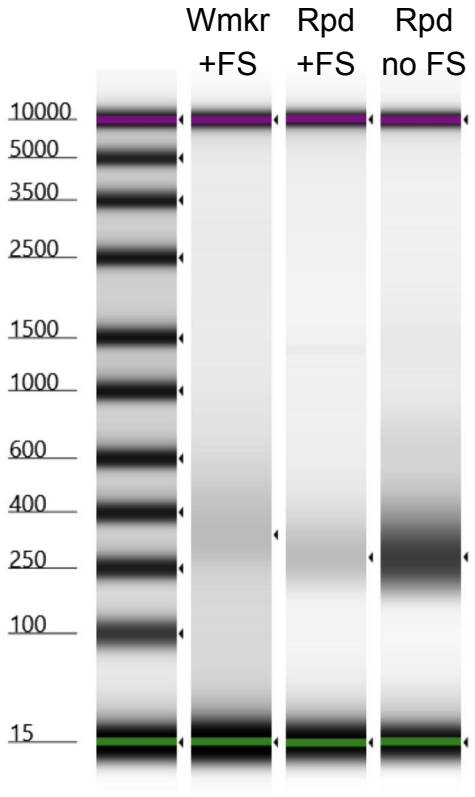


Watchmaker -rRNA
883,802 total reads



Rapid library insert sizes are smaller than Watchmaker, which matches Tapestation





Note from Illumina re library size:

The smaller library sizes (~275bp) are expected with the Rapid kit.

The insert/library size is dictated by the type of tagmentation beads used (eBLTS) which have higher density transposomes than BLTL beads.

This size is consistently reproducible with the kit, regardless of sample type, which is potentially a big advantage.

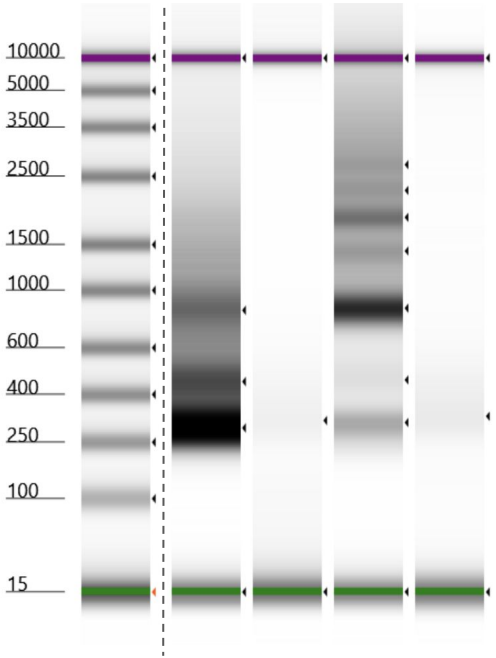
From our pilot, it does not seem that the library size affects downstream analysis or results.

Bonus clinical case

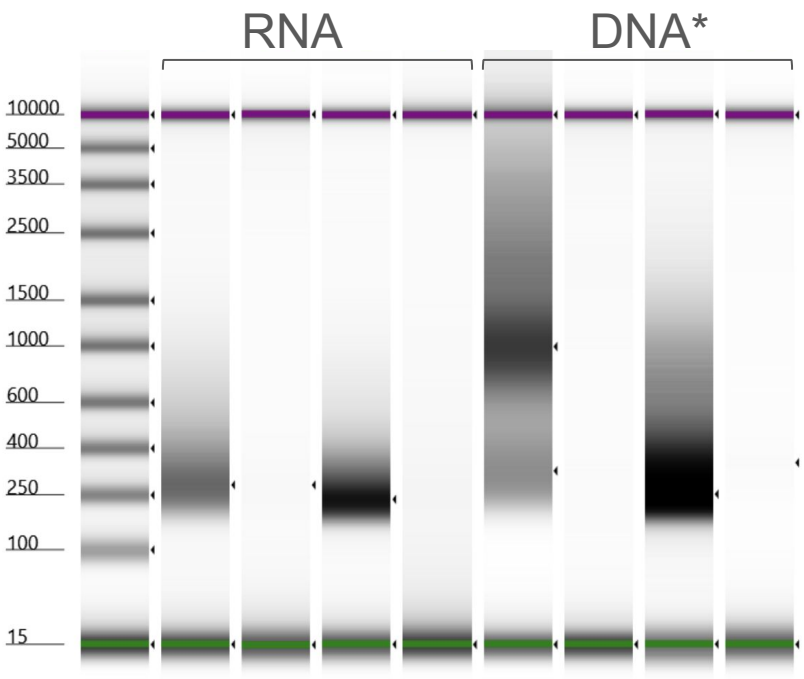
CSF and Plasma

RR242 libraries

Watchmaker

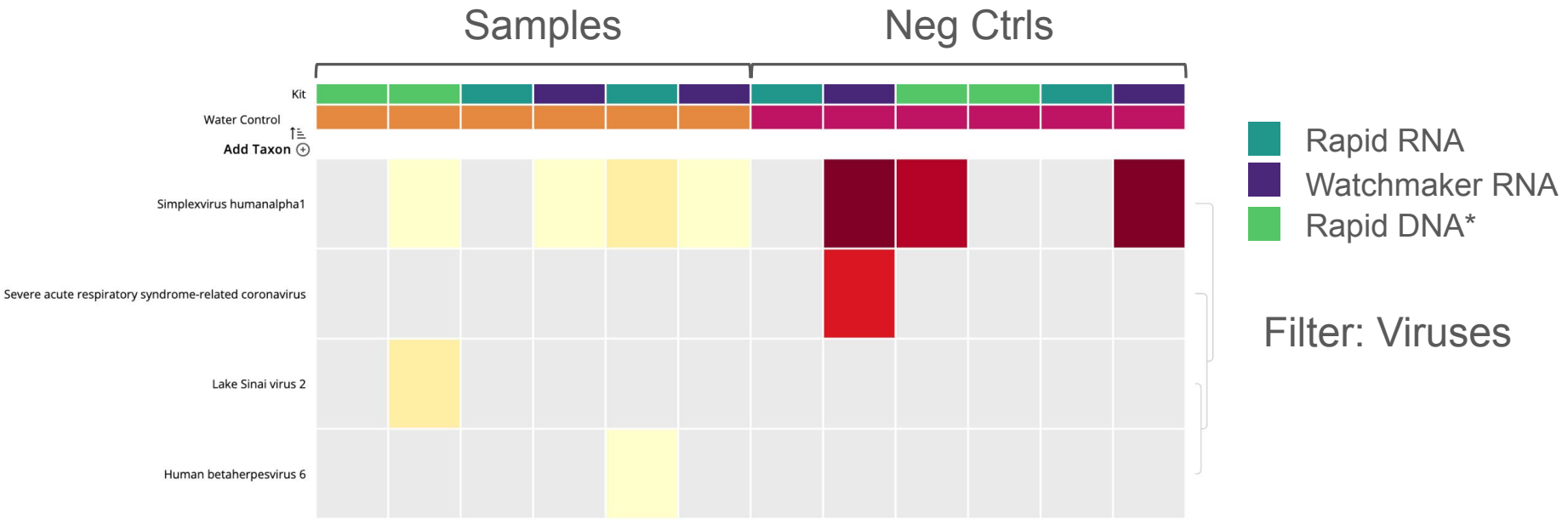


Illumina Rapid



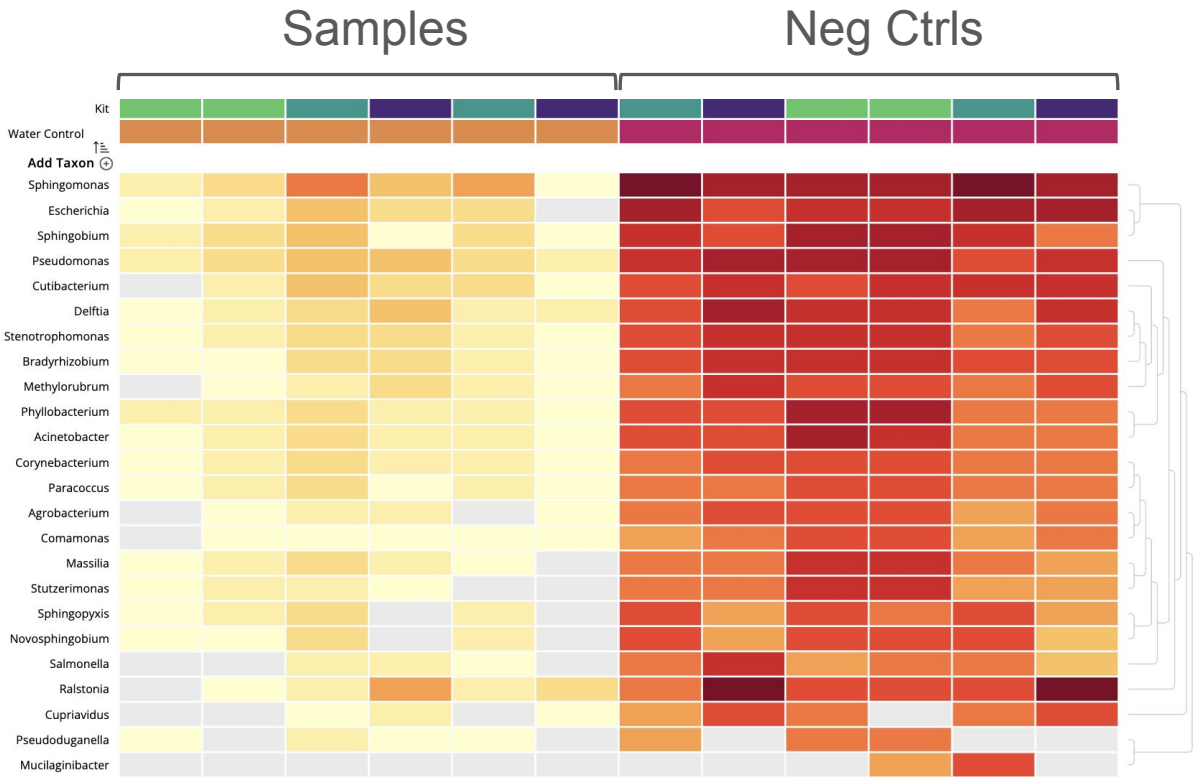
* Note: Illumina Rapid RNA kit is not designed for DNA input

No viruses detected above background in samples



* Note: Illumina Rapid RNA kit is not designed for DNA input

All bacteria have higher rPM in neg ctrls than samples



Rapid RNA

Watchmaker RNA

Rapid DNA*

Filter: Bacteria

* Note: Illumina Rapid RNA kit is not designed for DNA input

Summary

- Illumina Rapid RNA kit is really fast and easy
- Pathogen results from Rapid kit are comparable to Watchmaker
  - Low input samples
  - Nasal swabs, CSF, plasma
  - No concern about false positives or false negatives, compared to Watchmaker
- Human transcriptome was not evaluated in this pilot
- Insert sizes are smaller with the Rapid kit
  - This is expected for the Rapid kit and is likely an advantage - uniform libraries (~275bp) are expected for all samples regardless of type
- The Rapid kit is a very good starter kit for groups new to library preparation